Structured Nonsmooth Optimization (II)

Discussion with Prof. Anthony Man-Cho So

Presenter: Luxuan Li

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Nonsmooth DC optimization[1]

Consider.

$$\min_x \; F(x) = \varphi(x) - g(x), \qquad \text{(P)}$$

- φ weakly convex, g convex; both may be nonsmooth;
- hence F may be nonsmooth and nonconvex;
- **focus:** finite-time complexity analysis when g is nonsmooth.

Motivating example. Nonconvex regularized problem

$$\min_x \; F(x) = \underbrace{f(x) + h(x)}_{\varphi(x)} - g(x),$$

where f is L-smooth and $h - g$ is a nonconvex regularizer (e.g., capped-ℓ_1) admitting an efficient DC decomposition.

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[1] L. Li, C. Cui, and X. Wang. *MoSSP: A Momentum-Based Single-Loop Stochastic Penalty Method for Nonconvex Constrained DC-Regularized Optimization*. Accepted at ICML, 2026.

Stationarity in Nonsmooth (DC) Optimization

- **Clarke stationarity** (conventional target): one seeks

$$0 \in \partial_C F(\bar{x}).$$

- **Critical point in DC optimization:** $\bar{x}$ is a critical point of Problem (P) if

$$0 \in \partial\varphi(\bar{x}) - \partial g(\bar{x}), \quad \text{equivalently,} \quad \partial\varphi(\bar{x}) \cap \partial g(\bar{x}) \neq \emptyset.$$

- **Relation:** weaker

$$\partial_C(\varphi - g)(x) \subseteq \partial\varphi(x) - \partial g(x).$$

Asymptotic Stationarity vs. Finite-Time Convergence

For locally Lipschitz F, a typical asymptotic convergence proof gives

$$x^k \to x^\star, \qquad 0 \in \partial_C F(x^\star),$$

but *not* $\operatorname{dist}(0, \partial_C F(x^k)) \to 0$, and no finite $K(\varepsilon)$ with $\operatorname{dist}(0, \partial_C F(x^K)) \le \varepsilon$ is guaranteed to exist.

Key reason

At a nonsmooth point, the subdifferential set may expand abruptly:
$F(x) = |x|$ has $0 \in \partial_C F(0) = [-1, 1]$, yet $\operatorname{dist}\big(0, \partial_C F(x)\big) = 1$ for all $x \neq 0$.

Finite-time Convergence Criteria in DC optimization.

- **Critical point:** $0 \in \partial\varphi(\bar{x}) - \partial g(\bar{x})$;
- **Finite-time relaxation:** Given $\varepsilon > 0$, *exact* same-point ε-criticality means

$$\operatorname{dist}(0, \partial\varphi(\bar{x}) - \partial g(\bar{x})) \le \varepsilon.$$

- **Relation:** $\operatorname{dist}(0, \partial\phi - \partial g) \le \operatorname{dist}(0, \partial_C(\phi - g))$.

Question 1. Can any finite-time algorithm return an *exact* same-point ε-critical point of Problem (P)?

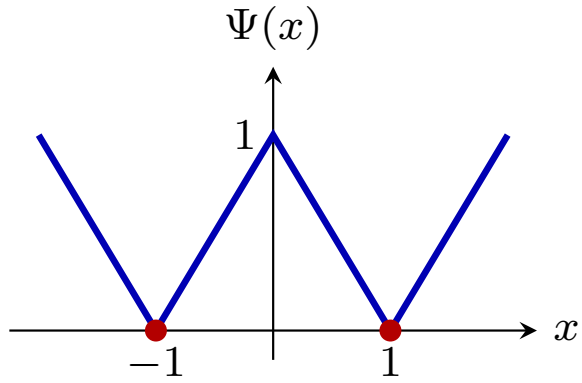
The Same Obstruction in Nonsmooth DC Optimization

A DC example

$$\Psi(x) = \big||x| - 1\big| = \phi(x) - g(x),$$

with

$$\phi(x) = 3 + 4(|x| - 1)_+,$$
$$g(x) = |x + 1| + |x| + |x - 1|.$$



- $x^* = 1$ is a global minimizer;
- For the sequence $x_k = 1 - \frac{1}{k+2}$, $\forall k \geq 1$:

  $$\text{(Asymptotic)} \quad x_k \to 1, \quad 0 \in \partial_C(\phi - g)(1) \subseteq \partial\phi(1) - \partial g(1),$$

  yet every finite iterate satisfies

  $$\text{(Finite-time)} \quad \operatorname{dist}\big(0, \partial\phi(x_k) - \partial g(x_k)\big) = 1 > \varepsilon.$$

- The criticality residual fails to vanish in finite time.

Consequence

Convergence to a exact same-point critical point does not ensure that the residual becomes small along the trajectory.

Unattainability of the ε-Critical Point

For general nonsmooth nonconvex optimization, Zhang et al.[2] showed that ε-Clarke stationarity is not finitely attainable.

In nonsmooth DC optimization this obstruction is *not* inherited automatically:

- the DC oracle is *stronger*: it may reveal $\partial\phi(x)$ and $\partial g(x)$ separately;
- the DC residual is *weaker*: $\operatorname{dist}(0, \partial\phi - \partial g) \leq \operatorname{dist}(0, \partial_C(\phi - g))$.

[2] J. Zhang, H. Lin, S. Jegelka, S. Sra, and A. Jadbabaie, *Complexity of Finding Stationary Points of Nonsmooth Nonconvex Functions*, ICML, 2020, Theorem 5.

Unattainability of the ε-Critical Point

To answer Question 1, we offer two approaches to hard instances built with GPT's assistance.

Theorem A. Fix $\Delta, L, G > 0$. There is an constant $c_0 > 0$ such that, for every finite K, every $\varepsilon \in \big(0,\, c_0 \min\{L, G\}\big]$, and every (possibly randomized) algorithm $\mathcal{A}$ interacting with $\mathcal{O}_{\mathrm{DC}}$, there exists $(\phi, g) \in \mathcal{D}(\Delta, L, G)$ such that, w.p. $\geq 1/2$, the first K query points of $\mathcal{A}$ satisfy

$$\operatorname{dist}\big(0, \partial\phi(x^k) - \partial g(x^k)\big) > \varepsilon, \qquad k = 1, \ldots, K.$$

Moreover, for this instance there exists a DC-critical point $z^\star$ that is not Clarke-stationary:

$$0 \in \partial\phi(z^\star) - \partial g(z^\star), \qquad 0 \notin \partial_C(\phi - g)(z^\star).$$

Unattainability of the ε-Critical Point

Proof skeleton (barrier by reduction from the convex small-subgradient LB[3]):

1. **Convex LB.** For 1-D convex 1-Lipschitz q, no finite-time first-order algorithm guarantees $\operatorname{dist}(0, \partial q(x)) \le \varepsilon$.
2. **2-D lifting.**

$$\phi_q(x,y) = q(x) + |y| + |y-1|, \quad g_q(x,y) = |y-1|, \quad F_q = \phi_q - g_q = q(x) + |y|.$$

3. **Reduction validity.** $D(x,y) := \operatorname{dist}(0, \partial\phi_q - \partial g_q) \ge \operatorname{dist}(0, \partial q(x))$;
4. **Strict separation.** At $(x^\star, 1)$, $x^\star \in \arg\min q$: $0 \in \partial\phi_q - \partial g_q$ but $\partial_C F_q(x^\star, 1) = \partial q(x^\star) \times \{1\} \not\ni 0$.
5. **Parameters.** Fixed $(\Delta, L, G) = (1, \sqrt{2}, \sqrt{5})$, K-independent; rescale by $\rho = \frac{1}{3}\min\{L, G\}$ to arbitrary (Δ, L, G).

Limitation. The hardness is inherited from a one-dimensional convex lower bound;

[3] G. Kornowski and O. Shamir, *On the Complexity of Finding Small Subgradients in Nonsmooth Optimization*, NeurIPS 2022 Workshop; arXiv:2209.10346v2.

Unattainability of the ε-Critical Point

Theorem B. Fix $\Delta, L, G > 0$. There exists an *absolute* constant $c > 0$ such that, for every $K \le cG/L$, every $\varepsilon \in (0, L)$, and every (possibly randomized) algorithm $\mathcal{A}$ interacting with $\mathcal{O}_{\mathrm{DC}}$, there exists $(\phi, g) \in \mathcal{D}(\Delta, L, G)$ such that, w.p. $\ge 1/2$, the first K query points of $\mathcal{A}$ satisfy

$$\operatorname{dist}\big(0, \partial\phi(x^k) - \partial g(x^k)\big) > \varepsilon, \qquad k = 1, \dots, K.$$

Moreover, for this instance there exists a DC-critical point s that is not Clarke-stationary:

$$0 \in \partial\phi(s) - \partial g(s), \qquad 0 \notin \partial_C(\phi - g)(s).$$

Answer 1. A traditional exact same-point ϵ-critical point cannot be obtained in finite time, even under the strong component-wise DC oracle.

Proof skeleton (DC resisting oracle, adapted from Zhang et al. 2020):

1. **Background.** (ϕ_0, g_0) with hidden kinks s.t.

$$\operatorname{dist}\big(0, \partial\phi_0(x^k) - \partial g_0(x^k)\big) > \varepsilon, \quad \forall\, k.$$

2. **Graft.** $c_s(x) = M|x - s| + \beta - Ms$; $\phi_s = \phi_0 + c_s$, $g_s = g_0 + c_s$. Two s-invariants:

$$F_s := \phi_s - g_s = \phi_0 - g_0, \qquad c_s(x) = \beta - Mx \text{ on } \{x < s\}.$$

3. **Indistinguishability.** For every admissible $s > \max_k x^k$, the transcript $(\phi_s, g_s, \partial\phi_s, \partial g_s)\big|_{x^k}$ is s-invariant.
4. **Separation.** At $x = s$, $\partial c_s(s) = [-M, M]$, hence

$$0 \in \partial\phi_s(s) - \partial g_s(s), \qquad 0 \notin \partial_C F_s(s).$$

Two admissible $s \neq s'$ share the same transcript $\Rightarrow \Pr[\text{fail}] \geq \frac{1}{2}$.

Limitation: requires $K \leq cG/L$ (equiv. $G_K = O(KL)$);

A Unified DC Criticality

Question 2. Can we define a novel DC criticality which is *finite-time verifiable* under the component-wise oracle $\mathcal{O}_{\mathrm{DC}}$, yet keeps the optimality meaning?

Tools.

- The convex δ-subdifferential.

  For convex $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, $\delta \geq 0$ and $\bar{x} \in \operatorname{dom} h$,

  $$\partial_\delta h(\bar{x}) := \left\{ v \in \mathbb{R}^n : \; h(z) \geq h(\bar{x}) + \langle v, z - \bar{x} \rangle - \delta \;\; \forall\, z \in \mathbb{R}^n \right\}, \;\; \partial_0 h(\bar{x}) = \partial h(\bar{x}).$$

- With the **Fenchel–Young gap** $\Phi_h(x, w) := h(x) + h^*(w) - \langle w, x \rangle \geq 0$:

  $$v \in \partial_\delta h(\bar{x}) \iff \Phi_h(\bar{x}, v) \leq \delta.$$

Design requirements.

- **finite-time computable.**
- **optimality meaning.**
- **asymptotic exactness.**
- **unification.**

Answer 2: The $(\delta_\phi, \delta_g, \varepsilon)$-Critical Point

Definition. Given $\delta_\phi, \varepsilon \geq 0$ and $\delta_g > 0$, a point $\bar{x} \in \operatorname{dom}\phi \cap \operatorname{dom} g$ is called a $(\delta_\phi, \delta_g, \varepsilon)$-*critical point* of Problem (P) if

$$\operatorname{dist}\big(\partial_{\delta_\phi}\phi(\bar{x}),\, \partial_{\delta_g} g(\bar{x})\big) \leq \varepsilon.$$

Equivalent certificate. There exist $u, v \in \mathbb{R}^n$ with

$$\Phi_\phi(\bar{x}, u) \leq \delta_\phi, \qquad \Phi_g(\bar{x}, v) \leq \delta_g, \qquad \|u - v\| \leq \varepsilon.$$

- **same point:** both approximate subgradients at $\bar{x}$;
- $(0,0,0)$-critical $\iff$ $0 \in \partial\phi(\bar{x}) - \partial g(\bar{x})$ (exact DC criticality).

The Three Parameters: What Do They Measure?

Recall the definition. $(\delta_\phi, \delta_g, \varepsilon)$-critical means: $\exists u, v$ with

$$\Phi_\phi(\bar{x}, u) \le \delta_\phi, \qquad \Phi_g(\bar{x}, v) \le \delta_g, \qquad \|u - v\| \le \varepsilon.$$

The three parameters separate independent sources of approximation error.

- δ_ϕ**: subproblem solve accuracy.**

  How close did we get to the subproblem optimum?

  In DCA: Since $\min_x\{\phi(x) - \langle v^k, x\rangle\} = -\phi^*(v^k)$, we have

  $$\Phi_\phi(x^{k+1}, v^k) = \big\{\phi(x^{k+1}) - \langle v^k, x^{k+1}\rangle\big\} - \min_x\big\{\phi(x) - \langle v^k, x\rangle\big\},$$

- δ_g**: nonsmooth transport cost.**

  How much Fenchel gap accumulates when using $v \in \partial g$ away from its anchor?

  In DCA: $v^k \in \partial g(x^k)$ moved to x^{k+1} gives $\Phi_g(x^{k+1}, v^k)$.

- ε**: dual vector mismatch.**

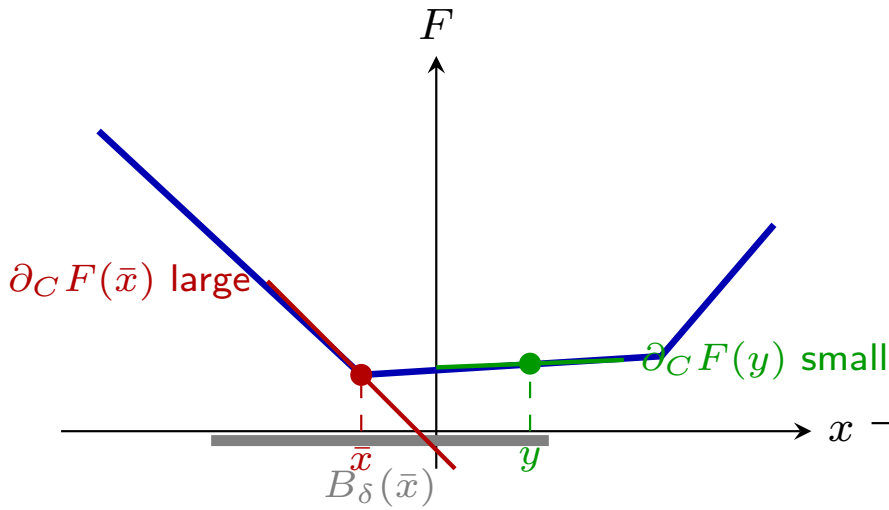
  How far apart are the two approximate subgradients?

  In DCA: at output $\bar{x}$, find $u \in \partial_{\delta_\phi}\phi(\bar{x})$ and $v \in \partial_{\delta_g} g(\bar{x})$, set $\varepsilon = \|u - v\|$.
  Exact DCA: $u = v = v^k$, $\varepsilon = 0$. Proximal DCA: $u^k = v^k - \gamma\Delta x^k$, $\varepsilon = \gamma\|\Delta x^k\|$.

Spatial Relaxation vs. Value Relaxation

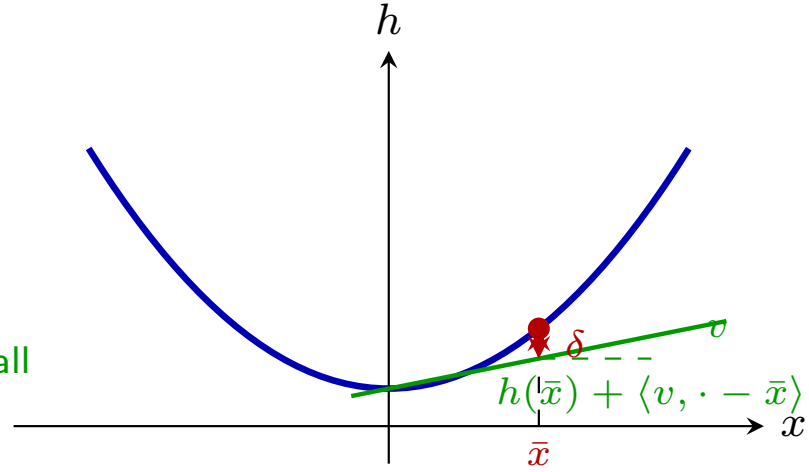
- (δ, ε)-**Goldstein stationary** [Goldstein 1977; Zhang et al. 2020]: $\operatorname{dist}(0, \partial_\delta^G F(z)) \le \varepsilon$ where $\partial_\delta^G F(z) := \operatorname{conv} \bigcup_{\|z'-z\| \le \delta} \partial_C F(z')$.
- **Ours**: both certificates stay at the same point $\bar{x}$.

Goldstein: spatial



Move to $y \in B_\delta(\bar{x})$ where gradient is small

Ours: value



Stay at $\bar{x}$, allow Fenchel gap $\le \delta$

Why value relaxation for DCA

- Goldstein stationarity: requires additional mechanisms (smoothing or neighborhood sampling);
- ✓ $(\delta_\phi, \delta_g, \varepsilon)$-**criticality**: uses quantities DCA already computes: $\Phi_\phi(x^{k+1}, v^k)$ and $\Phi_g(x^{k+1}, v^k)$.

Exact Consistency and Unification

I. Exact consistency. $(\delta_\phi, \delta_g, \varepsilon) \to 0$ recovers exact DC criticality $0 \in \partial\phi(\bar{x}) - \partial g(\bar{x})$.

II. Unification of existing relaxations.

- **Spatial notions** [Moudafi 2021; Sun 2023; Yao 2022]: offset $\|\bar{x} - \bar{y}\|$ becomes Fenchel slack $2G\|\bar{x} - \bar{y}\|$ (Lipschitz) or $\frac{L}{2}\|\bar{x} - \bar{y}\|^2$ (smooth);
- **Value-inexact DCA**: δ_ϕ *is* the subproblem gap — no conversion needed;
- **Goldstein stationarity** [Zhang et al. 2020]: (δ, ε)-Goldstein $\Rightarrow$ $(2G_\phi\delta, 2G_g\delta, \varepsilon)$-critical.

Complexity Recovered and Tightness

$\Delta_0 := F(x^0) - F^\star$.

method	**certificate after K iterations**	**complexity**
exact DCA	$(0, \frac{\Delta_0}{K}, 0)$	$K = O(\frac{\Delta_0}{\delta_g})$
inexact DCA	$(\sigma, \frac{\Delta_0}{K} + \sigma, 0)$	outer $O(\frac{\Delta_0}{\delta_g - \sigma})$
proximal DCA	$(0, \frac{\Delta_0}{K}, \sqrt{\frac{\Delta_0}{\gamma K}})$	$O(\max\{\frac{\Delta_0}{\delta_g}, \frac{\Delta_0}{\gamma \varepsilon^2}\})$
stochastic minibatch	$(O(\varepsilon^2), O(\varepsilon^2), \varepsilon)$	$O(\sigma^2 L_f \Delta_0 \, \varepsilon^{-4})$ SFO

Takeaways.

- all rows follow from one identity: the g-gap is part of each step's descent, hence $\delta_g \le \Delta_0/K$ after K steps;
- $\varepsilon = 0$ when one subgradient certifies both sides (exact DCA, $u = v = v^k$); prox shifts it by $\gamma\|\Delta x\|$; stochastic gradients add $\sqrt{\bar{V}}$;
- L_g-smooth g: $\delta_g = O(\varepsilon^2/L_g)$ recovers the classical ε-critical point, so $O(\varepsilon^{-2})$ cannot be improved;
- nonsmooth g: whether $O(\varepsilon^{-2})$ is optimal? open.

Contribution (to our knowledge). the first *non-asymptotic* complexity analysis of DCA-type methods for nonsmooth g that (i) does not smooth g and (ii) returns a optimality guarantee measure.

Discussion: Open Questions

- **lower bound, nonsmooth g:** are $\Omega(\varepsilon^{-2})$ component-wise queries necessary for a $(\delta_\phi, \delta_g, \varepsilon)$-critical point? can the hard-instance ideas of the barrier theorem be reused?
- **decomposition dependence:** the certificate depends on the split (ϕ, g), not only on F — acceptable, since DCA itself uses the split? is there a best split for a given budget?
- **meaning of ε:** it is 0, $\gamma\|\Delta x\|$, or $\sqrt{\bar{V}}$ depending on the method: does this distance have a geometric meaning in the dual space?

Summary

C1. **Unattainability of same-point ε-criticality.**

C2. **A same-point, three-parameter criticality definition with optimality guarantee.**

C3. **Unified Analysis framework for the whole DCA family.**

C4. **Complexity recovered without smoothing g.**